



Fig. 8: TopoR



Fig. 9: Electromagnetic crosstalks reduction in TopoR

tion costs. In comparison with other CAD systems, TopoR allows you to substantially reduce the overall wire length and the number of vias. This allows more space on a board. It also provides a possibility of increasing the clearance between wires and the size of contact pads. This helps to make the board smaller in size and reduce the number of layers.

TopoR offers the following key features:

High-quality autorouting. Autorouting quality is comparable to manual routing.

Improved electromagnetic com-

patibility. Wire parallelism is avoided considerably due to absence of preferred routing directions in layers. Effective usage of mounting surface and absence of preferred routing directions (not limited by 90° and 45°) result in reduction of electromagnetic crosstalks up to several times in comparison with other CAD systems.

Parallel optimisation of several layout versions. TopoR can perform parallel optimisation of several alternative topology variants, which differ in optimised parameters. It also allows users to select one or several variants.

Specifying minimum/preferred clearances for each net. Users can immediately define the technical possibility of routing the board within the limits of the specified number of layers, specified clearance and other technical limitations based on the routing density and the amount of technical constraints violations.

Displacing components on a routed board. TopoR allows users to move components on the routed board while preserving the connections topology. Geometry and positioning of components are automatically calculated to comply with the new components position. Movement of components, vias and wire branching is used to decrease wires' length while providing the utmost clearance space. Movement can be performed both manually and automatically, and the most effective wires shape is defined immediately.

Smart BGA component routing. Due to absence of preferred routing directions in layers and deep level of optimisation, TopoR provides quality routing of contemporary BGA components, which is traditionally very difficult with other routers.

To sum up, TopoR offers the following advantages:

1. PCB design time reduction. TopoR reduces PCB design time considerably in the following ways:

Routing of 100 per cent wires on a board within no time. TopoR offers routing of 100 per cent of wires on a board within a few seconds. Autorouting is followed by multiobjective optimisation by calculating different variants of laying wires. The routing process can also be stopped at any moment, and the engineer can immediately define the technical possibility of routing the board within the limits of the specified number of layers, clearance size and other technical limitations based on the routing density and the amount of technical constraints violations. This helps to avoid wait for many hours, which is typical with traditional autorouters.